

Lab 1 – RedFlag Product Description

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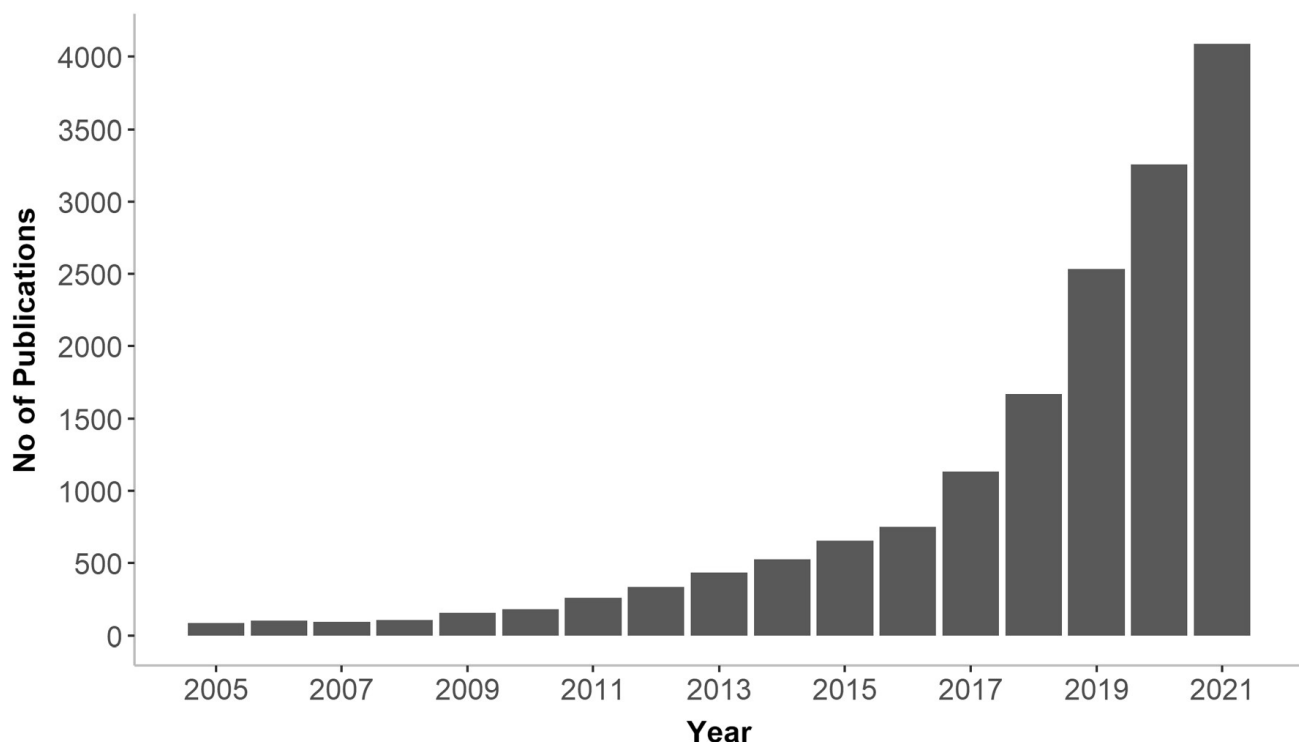
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1 Introduction

When over 75% of US Americans get at least some of their news online (Pew Research Center 2025), makes an impact whether that information is accurate. In a study done by the Alan Turing Institute (2023), over 90% of respondents indicated they had encountered misinformation online. While the exact amount is difficult to directly quantify, the correlation can be viewed in Figure 1 of how much research has been done on social media misinformation- proving that it is being noticed and worried about more and more.

Figure 1

Number of Publications about Online Misinformation by Year



Misinformation spreads faster than the truth (Vosoughi et al., 2017). Research indicates the two primary motivators of misinformation spread are a lack of attention from users (Ordway, 2021) and the platforms themselves encouraging this behavior (Allen, 2023). It is easiest and most rewarding to simply share what comes across ones feed without verifying.

Current fact-checking systems are ineffective. A few platforms have system-specific

checkers, like the community notes feature on X, but that won't help users on Instagram or news sites. People can manually verify facts themselves, but that not only requires increased effort on the part of the user- going to one or more separate sites, entering information, etc.- that they are unlikely to spend- but is actively discouraged by the platform, when sharing is as easy as a simple click. Even if the user does attempt to do so, accurately gauging the credibility of a source on one's own can be difficult. And when algorithms reinforce an existing bias- showing people things they are likely to be inclined to agree with, they are more likely to be fooled.

What is needed is a tool that is cross-platform, signals misinformation on the social media or other web page itself, can be interacted with and provides information without leaving the current site, and is based on a database that provides both credibility data and bias information and competing sources.

Here is where we introduce RedFlag.

2 Product Description

RedFlag is a fact-checking and bias indicator utility that works across multiple platforms. It runs in the background and scans text to find any inaccuracies and provide sources for the user to look into. This information and sources provided give the user a way to look further into the misinformation. There is also the BiasStar which will add a star to any politically divisive content and offers further viewpoints from sources across the political spectrum.

2.1 Key Product Features and Capabilities

Real-Time Integrated Detection, Feature: Scans the user's screen to find any misinformation, Significance: Works automatically in the background without requiring user input, Addresses: Tackles the inconvenience of the user having to leave their current browsing experience by bringing sources directly to them

Highlight Warnings, Feature: Highlights any questionable information on the users'

browsing experience with no user input, Significance: Does not intrude on the users' experience and presents the highlight subtly, Addresses: Attempts to intervene before the user shares misinformation, while not entirely interrupting the experience

Balanced Perspectives Engine (BiasStar), Feature: Identifies political content and provides multiple viewpoint sources across the political spectrum, Significance: Doesn't label ideologies explicitly but offers an evaluation from every perspective, Addresses: Tackles how social media may reinforce bias and echo chambers

On Demand Explanations, Feature: Detailed overviews are provided on the flagged information and explained to the user, Significance: Allows the user to make a more informed decision without dictating how the user should think on the topic, Addresses: Provides better credibility towards content and allows the user to think further on the information

2.2 Major Components (Hardware/Software)

Presentation Layer (Frontend), Components: The RedFlag browser extension, Function: Displays real time alerts and a view panel for users, Tech Stack: Built with Javascript for browser extensions

Application Layer (Backend), Components: Content detection, Misinformation Analysis, Fact Verification, Balanced Perspectives Engine, and Alert & Recommendation Generator, Tech Stack: Python (Flask/Django) for the information datasets

Data Layer (Database), Components: PostgreSQL database, Stored Data: Source Credibility Database (Includes reliability scores, and political alignments), User Interaction Data, and Content Intelligence Data (Known false claims, topic classifications)

External Interfaces / APIs, Fact checking API: Compares questionable information against the database, News API: Takes public news as input and categorizes by topic

LLM API: Parses database information to summarize its findings and increase fact-checking speed

3 Identification of Case Study

For whom is the product being developed? Why? Who else might use this in the future?

WHO: 16 year-old teenager, Daiskue, spends large amounts of his free time scrolling through his phone on Instagram PROBLEM: He recognizes some false information on his Instagram feed, but doesn't want to spend the time researching every post he comes across. FEATURES: Redflag misinformation indicators. SCENARIO: Daiskue enables Redflag on his browser, and indicators pop up on questionable text. He receives multiple sources of information on questionable topics, and is able to go to those sources immediately. FUTURE: Word of mouth, discussion of Redflag with his friends

Table 1

Table of Comparison Between RWP and Prototype

Feature	Prototype	Real World Product
a propulsion system	Rubber band	Belt
fixed position wheels	Same as RWP	Same as Prototype
variable position wheels	Same as RWP	Same as Prototype
cockpit	Simulated using PC-based display and Bluetooth	Operator will use
Exhaust distribution element	Simulated on PC-display	Real combustion

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3.1 Prototype Development Challenges

- Describe the expected challenges to be encountered while completing the prototype
– e.g., knowledge missing, capability missing, supporting technology issues.

Glossary

BiasStar

Credibility Score

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